

## Answers

### VARC

|              |              |              |              |                 |                 |              |              |
|--------------|--------------|--------------|--------------|-----------------|-----------------|--------------|--------------|
| 1. <b>B</b>  | 2. <b>C</b>  | 3. <b>A</b>  | 4. <b>C</b>  | 5. <b>B</b>     | 6. <b>A</b>     | 7. <b>C</b>  | 8. <b>A</b>  |
| 9. <b>C</b>  | 10. <b>D</b> | 11. <b>A</b> | 12. <b>A</b> | 13. <b>C</b>    | 14. <b>A</b>    | 15. <b>C</b> | 16. <b>A</b> |
| 17. <b>B</b> | 18. <b>D</b> | 19. <b>2</b> | 20. <b>3</b> | 21. <b>4132</b> | 22. <b>4312</b> | 23. <b>C</b> | 24. <b>C</b> |

### LRDI

|              |              |              |                |               |              |               |               |
|--------------|--------------|--------------|----------------|---------------|--------------|---------------|---------------|
| 25. <b>D</b> | 26. <b>C</b> | 27. <b>4</b> | 28. <b>9</b>   | 29. <b>5</b>  | 30. <b>B</b> | 31. <b>C</b>  | 32. <b>17</b> |
| 33. <b>C</b> | 34. <b>A</b> | 35. <b>B</b> | 36. <b>D</b>   | 37. <b>25</b> | 38. <b>D</b> | 39. <b>26</b> | 40. <b>C</b>  |
| 41. <b>B</b> | 42. <b>A</b> | 43. <b>6</b> | 44. <b>140</b> |               |              |               |               |

### Quant

|              |              |               |              |               |                |                |              |
|--------------|--------------|---------------|--------------|---------------|----------------|----------------|--------------|
| 45. <b>D</b> | 46. <b>B</b> | 47. <b>C</b>  | 48. <b>D</b> | 49. <b>15</b> | 50. <b>7</b>   | 51. <b>6</b>   | 52. <b>A</b> |
| 53. <b>B</b> | 54. <b>C</b> | 55. <b>C</b>  | 56. <b>D</b> | 57. <b>B</b>  | 58. <b>16</b>  | 59. <b>407</b> | 60. <b>7</b> |
| 61. <b>B</b> | 62. <b>C</b> | 63. <b>45</b> | 64. <b>B</b> | 65. <b>C</b>  | 66. <b>967</b> |                |              |

## Explanations

### VARC

#### 1. **B**

The passage suggests that historians can rely on disciplines such as archaeology, among others, to establish basic facts. The relevant portion of the passage is:

*"But [to] praise a historian for his accuracy is like praising an architect for using well-seasoned timber or properly mixed concrete in his building. It is a necessary condition of his work, but not his essential function. It is precisely for matters of this kind that the historian is entitled to rely on what have been called the 'auxiliary sciences' of history—archaeology, epigraphy, numismatics, chronology, and so forth."*

In this context, the "auxiliary sciences" are mentioned as tools that historians can use to ensure the accuracy of basic facts. Archaeology is included in this list, suggesting that it helps historians in ascertaining factual accuracy by providing evidence from material remains, artifacts, and other archaeological findings. Therefore, Option B correctly reflects the role of archaeology in supporting historians in their pursuit of factual accuracy.

[VIDEO SOLUTION](#)

#### 2. **C**

Option C is the correct answer because it aligns with the passage's perspective that the interpretation of facts is subjective and can be influenced by different perspectives. The passage argues that historians play a crucial role in selecting and interpreting facts, and Option C supports this idea by suggesting that facts, like truth, can be relative.

If facts are relative, it means that what one person considers a fact may not be viewed the same way by another person. This relativity of facts supports the notion that the historian's interpretation and perspective play a significant role in determining what is considered a fact. Therefore, Option C, if true, reinforces the passage's claim that facts are not entirely objective and independent of the historian's perspective, and it does not weaken the passage's argument.

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### 3. A

The passage suggests that while establishing basic facts is necessary, the essential function of historians goes beyond this. It emphasizes the selective and interpretive nature of historical writing, where historians are expected to go deeper into understanding the context and motivations behind historical events.

Option A, which involves exploring the socio-political and economic factors that led to the Battle of Hastings, is in line with the idea of providing a nuanced interpretation. This option indicates a focus on understanding the underlying causes and influences that shaped the historical event, reflecting a more comprehensive and contextual approach to historical writing.

Option B: While timelines are important, the author suggests that the historian's essential function goes beyond establishing basic chronological facts.

Option C: The author acknowledges the importance of basic facts but argues that historians must go beyond mere fact-finding.

Option D: While the author acknowledges the use of auxiliary sciences, it also suggests that the historian's focus should go beyond relying solely on these sciences for basic facts, emphasizing the historian's selective and interpretive role in presenting historical events.

VIDEO SOLUTION

### 4. C

*"According to the common-sense view, there are certain basic facts which are the same for all historians and which form, so to speak, the backbone of history:*

The common-sense view of history, as described in the passage, emphasizes that history consists of a body of ascertained facts found in documents and engravings. In contrast, Option C suggests that history is like science, framing it as a selective system of cognitive orientations to reality. This characterization does not align with the common-sense view, which is more concerned with factual accuracy and tangible historical evidence rather than presenting history as a selective system akin to science. Therefore Option C is the correct answer.

Option A accurately reflects the common-sense view as described in the passage.

The common-sense view includes a broader acceptance of historical methods beyond positivism, as mentioned in the passage (option B) and as per the passage, involves a fallacy in believing that historical facts are objective and independent of interpretation (Option D).

VIDEO SOLUTION

### 5. B

Option B is not supported by the information in the passage. While the passage mentions that Netflix has offices across Europe, it also notes that the big decisions still rest with American executives. The passage suggests that Netflix's content might still exhibit a somewhat mid-Atlantic quality, and the company's executive decisions remain under the control of Americans, making it less accurate to claim that Netflix has completely transformed into a truly European entity.

Option A is supported by the passage, which mentions shows like "Lupin," a French crime caper on Netflix, becoming global hits.

Option C: The passage specifically mentions that, according to Ampere, a media-analysis company, in 2015, about 75% of Netflix's original content was American, but now the figure is half. This indicates a shift in the geographical distribution of Netflix's original programming, with a decrease in the proportion of American content.

Option D is true as the passage mentions that streaming services like Netflix account for about a third of all viewing hours, challenging the dominance of national broadcasters.

VIDEO SOLUTION



### 6. A

Option A is the correct answer because the passage emphasizes that the rise of Netflix in Europe is seen as a unifying force. The author notes that Netflix and similar streaming services, by pumping the same content into homes across the continent, contribute to making culture a cross-border endeavor. This is described as a shared experience among Europeans, as they binge-watch the same series. The idea is that having a common cultural experience, facilitated by platforms like Netflix, can be a unifying factor among the diverse populations of Europe. Therefore, the rise of Netflix is portrayed in a positive light as a force that brings people together through shared cultural consumption.

VIDEO SOLUTION

### 7. C

The passage mentions that certain genres, particularly murder mysteries and dramatic conflicts like "bloodthirsty maelstroms between arch Romans and uppity tribesmen," have a more universal appeal. This suggests that themes involving suspense, mystery, and conflicts can transcend cultural boundaries and be attractive to a broader audience.

Therefore, a murder mystery drama set in North Africa and France aligns with the passage's implication that such themes have a more universal appeal, making it likely to be successful with audiences across the diverse countries of the EU. So Option C is the correct answer.

Option A is incorrect as the passage states that comedy does not travel well.

Option B: While science fiction may have a global appeal, the passage emphasizes genres like murder mysteries having a more universal appeal.

Option D: The passage doesn't discuss the appeal of romantic dramas, and the trans-Atlantic setting may not necessarily align with the passage's suggestion that certain themes work better across borders within Europe.

VIDEO SOLUTION

### 8. A

The author concludes that the rise of Netflix in Europe is seen as a unifying force, emphasizing shared experiences through common series. If research were to show a wide variance in the popularity and viewing of Netflix shows across different EU countries (Option A), it would suggest that the impact of Netflix on cultural unity is not as consistent or unifying as the author implies. The wide variance could indicate that cultural preferences or barriers within different EU countries limit the effectiveness of Netflix as a unifying force across the entire region. Therefore Option A is the correct answer.

Option B: While this finding could have implications for Netflix's business, it doesn't necessarily address the cultural unifying aspect mentioned in the author's conclusion.

Option C: This finding doesn't directly relate to the author's conclusion about Netflix serving as a unifying force within Europe.

Option D: While this finding highlights audience preferences, it doesn't directly address the author's conclusion regarding the cross-border unifying role of Netflix in Europe.

VIDEO SOLUTION

9. C

The central idea of the passage is the promotion of sustainable shopping practices, particularly second-hand shopping, as a means to combat the detrimental environmental effects of the fashion industry. But, the passage also discusses the need for consumers to be mindful of the environmental impact of their clothing choices, **opting for high-quality items** that last longer and shed fewer microfibers.

The passage argues that opting for second clothing might not always be beneficial for the environment by highlighting the microfibre pollution that they can potentially cause. Now, if the second-hand clothes being sold were only of higher quality, it would take care of this problem (*They would be well advised to buy high-quality items that shed less and last longer [as this] combats both microfibre pollution and excess garments ending up in landfills*)

So, the correct answer is Option C.

Option A is more about the purchasing channel than the nature of the clothes so it does not necessarily undermine the central idea of the passage.

Option B supports the central idea by reducing environmental harm.

Option D could align with the sustainability goal and support the central idea, so it doesn't necessarily undermine it.

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10. D

Option D is the correct answer because the passage does not mention or suggest that the British don't buy second-hand clothing. Instead, the passage discusses challenges related to luxury brands and their reluctance to circulate their latest season stock globally at a cheaper price. The reasons mentioned include the financial aspect (Option A), concerns about brand image (Option B), and the desire to avoid devaluing their products (Option C). Therefore, the passage does not attribute the slow adoption of companies like ThredUP in the UK to the British not buying second-hand clothing.

VIDEO SOLUTION

## CAT Percentile Predictor



11. A

Option A is the correct answer because the passage emphasizes the environmental issues associated with fast fashion, including the wasteful disposal of garments in landfills. The opposite of this disposable and rapid turnover nature of fast fashion would be a more sustainable and durable approach, which aligns with the idea of "slow fashion."

The passage suggests that buying high-quality items that last longer is a way to combat the negative environmental impact of the fashion industry. Therefore, 'slow fashion' can be inferred to refer to clothes that are of high quality and long-lasting, promoting a more sustainable and environmentally friendly approach to fashion consumption.

VIDEO SOLUTION

12. A

The irony of "thrifting," as discussed in the passage, is rooted in its unintended environmental consequences. While thrifting is commonly associated with sustainable and eco-friendly practices, the passage highlights a potential environmental issue linked to the act. Specifically, the passage mentions a study commissioned by Patagonia that reveals older clothes, often found in second-hand stores, tend to shed more microfibers. These microfibers can end up in rivers and oceans, contributing to microfiber pollution. Therefore, the seemingly environmentally conscious act of thrifting, aimed at reducing waste, may inadvertently result in environmental problems through the shedding of microfibers during the washing of older garments. Therefore Option A is the correct answer.

VIDEO SOLUTION

13. C

Option C is the correct answer because the mention of "the Davos elite" in the passage serves to illustrate the perceived hypocrisy of wealthy individuals who profess to adhere to liberal values while simultaneously amassing the majority of the wealth. The author points to the contradiction between the elite's participation in events like discussions on creating a shared future and their exclusive access to privileges, symbolized by the use of private planes. This highlights the critique that the liberal rich, represented by the Davos elite, may not align their actions with the egalitarian ideals they claim to support, emphasizing a discrepancy between rhetoric and behavior.

Option A is incorrect because the passage does not explicitly connect the decline in liberal values to the rich benefiting, but rather to internal contradictions and hubris.

Option B is incorrect as the passage does not focus on how the debate around liberalism is captured by the rich; instead, it critiques the actions of the Davos elite.

Option D is incorrect because the passage does not suggest that the rise in liberalism has led to greater interest in shared futures from unlikely social classes; rather, it critiques the behavior of the Davos elite as hypocritical.

VIDEO SOLUTION

14. A

All the options, other than A, are direct signs of declining or ineffective liberalism.

Option B: *Creation of business aristocracy*, the author in the first paragraph says that liberalism promoted a meritocratic aristocracy and then went ahead to argue why the meritocratic aristocracy is not a good replacement of the old aristocracy. Creation of a business aristocracy and the rise of vast companies are against the ideals of liberalism.

Option C: *Democracy has degenerated into a theatre of the absurd*, this clearly shows the non-functionality of liberalism and is a pretty valid argument for the decline of liberalism.

Option D: *The gap between liberalism's claims about itself and the lived reality of the citizenry' is now so wide that 'the lie can no longer be accepted*, this line says that the gap between what liberalism asked us to do and what is actually different are two very different things. This too can be an evidence of liberalism's decline.

Option A: *And technological advances are reducing ever more areas of work into meaningless drudgery*, while this line does talk about the technological advancement in a negative sense, it does not necessarily provide evidence of the decline of liberalism per se. Instead, it highlights a potential consequence or critique within the context of technological advances. The negative impact of technology on certain types of work might be seen as a challenge that needs to be addressed within the liberal framework rather than direct evidence of the decline of liberalism.

The same challenge could be seen at a time when liberalism was prospering and thus is not an evidence of its decline.

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15.C

The author is likely to agree with the statement in Option C, as it aligns with the author's argument in the passage that liberalism has historically reformed itself in the face of challenges. The author emphasizes the ability of liberalism to address internal problems and reform rather than attributing its success to being the dominant ideal in the past century.

Option A: While the author emphasizes the historical ability of liberalism to reform itself, there is an acknowledgment of its current challenges and failures. Therefore, the author might not fully endorse the idea that claims about liberalism's disintegration are merely exaggerated misunderstandings.

Option B: The author may disagree with this statement as it suggests accepting liberalism's decline and seeking a substitute, which contradicts the call to reform liberalism.

Option D: The author may disagree with this statement as it simplifies the essence of liberalism, which the author argues encompasses various intellectual traditions and responses to the trade-off between rights and responsibilities.

[VIDEO SOLUTION](#)



16.A

Option A is the correct answer because the passage does not explicitly identify Deneen's repeated emphasis on premodern notions of liberty as a reason for faulting his conclusions. While the passage criticizes Deneen for his extreme pessimism about the future of liberalism, his narrow definition of liberalism limited to individual freedoms, and his fixation on the essence of liberalism, it does not specifically address his tendency to hark back to premodern notions. Therefore, Option A stands out as an exception, as it does not align with the explicitly stated reasons for faulting Deneen's conclusions in the passage.

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17.B

The sentence best fits in Blank 2 because it directly elaborates on the concept of 'productive dualism' introduced in before Blank 2. It provides additional information about the distinction between the "modern" and "traditional" sectors in poor countries' economies and highlights the historical perspective that contrasts developing and developed countries. This sentence helps set the stage for the subsequent discussion about the relevance of this distinction in the present context.

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18.D

The sentence best fits in Blank 2. Placing it there provides additional information about the timeline and challenges the previously acknowledged timeline in the scientific community, creating a logical flow in the paragraph.

The sentence does not fit in Blank 1 as it does not add anything to the Sentence preceding Blank 1. Placing the sentence in Blank 3 would disrupt the flow of the passage as there is a direct link between the sentence preceding and the sentence following Blank 3. (*script on clay tablets.....Many thousands of these clay tablets*.)

Similarly placing the Sentence in Blank 4 would disrupt the flow of the passage.

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19.2

Sentence 2 is the odd one out because it introduces a broader statement about attempts to ban books and silence authors, while the other sentences are specifically focused on the banning and challenging of books in the US. Sentences 1, 3, 4, and 5 collectively form a coherent paragraph discussing the banning of Northern Lights and the escalation of book challenges in the US, while Sentence 2 introduces a different perspective that doesn't directly contribute to the flow of the paragraph regarding the specific incidents and trends mentioned in the other sentences.

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20.3

Sentence 3 is the odd one out because it introduces a different perspective. While the other sentences discuss the negative outcomes associated with self-care in children, Sentence 3 suggests a need to consider that the bulk of research has yet to track long-term consequences. This sentence introduces a more neutral or balanced viewpoint that doesn't align with the general theme of the other sentences, which focus on the negative aspects of self-care for children.

The remaining sentences (1, 2, 4, and 5) can be put together to form a coherent paragraph discussing the concept of self-care in relation to children, particularly those referred to as "Latchkey children."



21. 4132

The correct order is 4-1-3-2.

Sentence 4 introduces the main idea that human consciousness is often referred to as an emergent property of the human brain. This gives context for the discussion to be followed.

Now if we consider Sentences 1 and 3 we can see that Sentence 1 ends with *"no single neuron holds complex information like self-awareness, hope or pride."* Sentence 3 builds on the idea in Sentence 1 by stating that, nonetheless, the collective activity of all neurons in the nervous system generates complex human emotions like fear and joy *"Nonetheless, the sum of all neurons ....."*. So we can infer that Sentence 3 must be following Sentence 1. Sentence 3 also supports the notion that emergent properties arise from the interaction of individual components.

Finally, Sentence 2 concludes the paragraph by explaining that although the exact mechanism of emergence in the human brain is not fully understood, neurobiologists agree that complex interconnections among the parts give rise to qualities specific to the whole. This sentence wraps up the discussion and reinforces the concept introduced Sentence 4.

Therefore the correct order is 4-1-3-2.

22. 4312

The correct order is 4-3-1-2.

Sentence 4 introduces the conflict between Western Christian concepts and indigenous African tradition. This gives context for the discussion to be followed.

Now we can see that Sentence 3 starts with *"However...."* which implies that this sentence is challenging the views presented in Sentence 4 by asserting the existence of a rich indigenous African cultural ethos. Therefore Sentence 3 must be following Sentence 4.

Sentence 2 introduces Chukwuemeka Ike and provides additional information about Chukwuemeka Ike's exploration of the conflict, adding depth to the discussion. Sentence 1 introduces an example of contemporary African writing that explores the conflict. It makes sense that first we give an example and then elaborate on it. Sentence 1 introduces the book; it is a logical inferences that Ike must be the author of this book.

Therefore Sentence 2 must be following Sentence 1.

Therefore the correct order is 4-3-1-2.

23. C

The passage discusses the phenomenon of counterfactual thinking, highlighting that people spontaneously create counterfactual alternatives to reality for various reasons. These reasons include explaining the past, preparing for the future, implicating various relations (including causal ones), affecting emotions, and supporting moral judgments. Additionally, the passage mentions that the ability to create counterfactuals develops throughout childhood and contributes to reasoning about other people's beliefs. Option C effectively encompasses these key points, making it the most accurate summary of the passage.

Option A focuses primarily on the preparation for the future aspect, neglecting the broader reasons for creating counterfactual alternatives.

Option B does not emphasize the developmental aspect and various reasons for creating

Option D inaccurately suggests that counterfactual thinking helps reverse past and future actions, which is not the main point of the passage, and it oversimplifies the role of counterfactuals.

24. C

The passage discusses the increasing frequency and intensity of heatwaves due to climate change, with vulnerable groups experiencing uneven impacts. It emphasizes that adaptation to heatwaves is a significant public policy concern. The research findings suggest that even vulnerable individuals may not perceive themselves as at risk of extreme heat, highlighting the importance of understanding how extreme heat is narrated. The passage specifically mentions the central role of the news media in warning people about the potential danger of heatwaves and their impacts on infrastructure and society. Option C effectively conveys the primary focus on heatwaves posing a substantial risk and the critical role of the media in alerting the public to this danger.

Option A implies a general importance of protection without specifically highlighting the role of the media in alerting people to the risks of heatwaves.

Option B acknowledges the vulnerability to heatwaves but it does not emphasize the role of the media in alerting people and suggests a broader critique of measures taken.

Option D mentions the need for news stories to become more effective but does not emphasize the central role of the media in alerting people to the risk of heatwaves, as the passage does.

LRDI

25. D

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 (1 + 1 + 1) to 27 (9 + 9 + 9)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column =  $45/3 = 15$  [The total number of coins in each row is the same. The total number of coins in each column is also the same.] ==> (1)

Let us represent the final configuration of the sacks in boxes as follows:

|     | Table |     |     |
|-----|-------|-----|-----|
|     | C-1   | C-2 | C-3 |
| R-1 |       |     |     |
| R-2 |       |     |     |
| R-3 |       |     |     |

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also \* => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now  $c > 5$  &  $c + 8 + 9$  should be a multiple of 3 =>  $c = 7$  is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also \*\* => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also \* => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 =>  $c = 3$  for  $c + 18$  to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

|     | Table     |           |     |
|-----|-----------|-----------|-----|
|     | C-1       | C-2       | C-3 |
| R-1 |           | 3,9,9 (7) |     |
| R-2 | 1,2,9 (4) |           |     |
| R-3 | 7,8,9 (8) |           |     |

From (1), The average in bag (1,1) is  $15 - 4 - 8 = 3$ .

From (1), The average in bag (1,3) is  $15 - 3 - 7 = 5$ .

|     | Table     |           |         |
|-----|-----------|-----------|---------|
|     | C-1       | C-2       | C-3     |
| R-1 | Avg = 3   | 3,9,9 (7) | Avg = 5 |
| R-2 | 1,2,9 (4) |           |         |
| R-3 | 7,8,9 (8) |           |         |

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and \*\* => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself  $3*3 = 9$ )

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and \* => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become  $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 =>  $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) |           |           |
| R-3 | 7,8,9 (8) |           |           |

Consider bag (3,3)

0 sacks have more than 5 coins and \*\* => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now  $c = 1$  or 2 or 3 or 4 =>  $c = 1$  or 4 for number of coins to be a multiple of 3.

But  $c = 1$  as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only  $1*$

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | Avg = 2   | Avg = 9   |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only  $1*$  => smallest elements should be 1.

=> 1, b, c are the coins where  $b + c = 5$  and b, c can't be equal to 1 and less than 5 =>  $2 + 3 = 5$  is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and \*\* => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and  $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | 1,8,9 (6) | 1,1,1 (1) |

Sum of coins in 3rd row =  $8*3 + 6*3 + 1*3 = 45$ .

[VIDEO SOLUTION](#)



26. C

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

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=> The total number of coins in a box range from 3 (1 + 1 + 1) to 27 (9 + 9 + 9)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column =  $45/3 = 15$  [The total number of coins in each row is the same. The total number of coins in each column is also the same.] => (1)

Let us represent the final configuration of the sacks in boxes as follows:

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=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also \*\* => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also \* => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 =>  $c = 3$  for  $c + 18$  to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

|     | Table     |           |     |
|-----|-----------|-----------|-----|
|     | C-1       | C-2       | C-3 |
| R-1 |           | 3,9,9 (7) |     |
| R-2 | 1,2,9 (4) |           |     |
| R-3 | 7,8,9 (8) |           |     |

From (1), The average in bag (1,1) is  $15 - 4 - 8 = 3$ .

From (1), The average in bag (1,3) is  $15 - 3 - 7 = 5$ .

|     | Table     |           |         |
|-----|-----------|-----------|---------|
|     | C-1       | C-2       | C-3     |
| R-1 | Avg = 3   | 3,9,9 (7) | Avg = 5 |
| R-2 | 1,2,9 (4) |           |         |
| R-3 | 7,8,9 (8) |           |         |

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and \*\* => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself  $3 \times 3 = 9$ )

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and \* => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become  $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 =>  $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) |           |           |
| R-3 | 7,8,9 (8) |           |           |

Consider bag (3,3)

0 sacks have more than 5 coins and \*\* => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now  $c = 1$  or 2 or 3 or 4 =>  $c = 1$  or 4 for number of coins to be a multiple of 3.

But  $c = 1$  as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only  $1 \times$

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | Avg = 2   | Avg = 9   |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only  $1 \times$  => smallest elements should be 1.

=> 1, b, c are the coins where  $b + c = 5$  and b,c can't be equal to 1 and less than 5 =>  $2 + 3 = 5$  is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and \*\* => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and  $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | 1,8,9 (6) | 1,1,1 (1) |

Bags (2,1), (3,1), (1,2), (3,2), (2,3) have at least 1 sack with 9 coins. => Total of 5 bags.

[VIDEO SOLUTION](#)

## 27.4

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 ( $1 + 1 + 1$ ) to 27 ( $9 + 9 + 9$ )

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column =  $45/3 = 15$  [The total number of coins in each row is the same. The total number of coins in each column is also the same.] => (1)

Let us represent the final configuration of the sacks in boxes as follows:

|     | Table |     |     |
|-----|-------|-----|-----|
|     | C-1   | C-2 | C-3 |
| R-1 |       |     |     |
| R-2 |       |     |     |
| R-3 |       |     |     |

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also \* => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now  $c > 5$  &  $c + 8 + 9$  should be a multiple of 3 =>  $c = 7$  is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also \*\* => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also \* => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 =>  $c = 3$  for  $c + 18$  to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

|     |           | Table     |     |
|-----|-----------|-----------|-----|
|     | C-1       | C-2       | C-3 |
| R-1 |           | 3,9,9 (7) |     |
| R-2 | 1,2,9 (4) |           |     |
| R-3 | 7,8,9 (8) |           |     |

From (1), The average in bag (1,1) is  $15 - 4 - 8 = 3$ .

From (1), The average in bag (1,3) is  $15 - 3 - 7 = 5$ .

|     |           | Table     |         |
|-----|-----------|-----------|---------|
|     | C-1       | C-2       | C-3     |
| R-1 | Avg = 3   | 3,9,9 (7) | Avg = 5 |
| R-2 | 1,2,9 (4) |           |         |
| R-3 | 7,8,9 (8) |           |         |

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and \*\* => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself  $3 \times 3 = 9$ )

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and \* => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become  $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 =>  $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

|     |           | Table     |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) |           |           |
| R-3 | 7,8,9 (8) |           |           |

Consider bag (3,3)

0 sacks have more than 5 coins and \*\* => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now  $c = 1$  or 2 or 3 or 4 =>  $c = 1$  or 4 for number of coins to be a multiple of 3.

But  $c = 1$  as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only  $1 \times$

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

|     |           | Table     |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | Avg = 2   | Avg = 9   |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only  $1 \times$  => smallest elements should be 1.

=> 1, b, c are the coins where  $b + c = 5$  and b,c can't be equal to 1 and less than 5 =>  $2 + 3 = 5$  is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

|     |           | Table     |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and \*\* => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and  $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

==> Final required table, bracket number => average coins per sack in the bag.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | 1,8,9 (6) | 1,1,1 (1) |

Average = Median in boxes (3,1), (2,2), (2,3) and (3,3) => 4 boxes.

VIDEO SOLUTION

28.9

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 (1 + 1 + 1) to 27 (9 + 9 + 9)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column =  $45/3 = 15$  [The total number of coins in each row is the same. The total number of coins in each column is also the same.] => (1)

Let us represent the final configuration of the sacks in boxes as follows:

|     | Table |     |     |
|-----|-------|-----|-----|
|     | C-1   | C-2 | C-3 |
| R-1 |       |     |     |
| R-2 |       |     |     |
| R-3 |       |     |     |

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also \* => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now  $c > 5$  &  $c + 8 + 9$  should be a multiple of 3 =>  $c = 7$  is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also \*\* => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also \* => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 =>  $c = 3$  for  $c + 18$  to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

|     | Table     |           |     |
|-----|-----------|-----------|-----|
|     | C-1       | C-2       | C-3 |
| R-1 |           | 3,9,9 (7) |     |
| R-2 | 1,2,9 (4) |           |     |
| R-3 | 7,8,9 (8) |           |     |

From (1), The average in bag (1,1) is  $15 - 4 - 8 = 3$ .

From (1), The average in bag (1,3) is  $15 - 3 - 7 = 5$ .

|     | Table     |           |         |
|-----|-----------|-----------|---------|
|     | C-1       | C-2       | C-3     |
| R-1 | Avg = 3   | 3,9,9 (7) | Avg = 5 |
| R-2 | 1,2,9 (4) |           |         |
| R-3 | 7,8,9 (8) |           |         |

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and \*\* => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself  $3*3 = 9$ )

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and \* => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become  $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 =>  $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) |           |           |
| R-3 | 7,8,9 (8) |           |           |

Consider bag (3,3)

0 sacks have more than 5 coins and \*\* => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now  $c = 1$  or  $2$  or  $3$  or  $4$  =>  $c = 1$  or  $4$  for number of coins to be a multiple of 3.

But  $c = 1$  as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1\*

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | Avg = 2   | Avg = 9   |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only 1\* => smallest elements should be 1.

=> 1, b, c are the coins where  $b + c = 5$  and b,c can't be equal to 1 and less than 5 =>  $2 + 3 = 5$  is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and \*\* => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and  $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

==> Final required table, bracket number => average coins per sack in the bag.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | 1,8,9 (6) | 1,1,1 (1) |

Bag (1,1) => 2 sacks with 1 coins, (2,1) => 1 sack, (2,2) => 1 sack, (3,2) => 1 sack, (1,3) => 1 sack, (3,3) => 3 sacks.

=> Total =  $2 + 1 + 1 + 1 + 1 + 3 = 9$  sacks.

[VIDEO SOLUTION](#)

29.5

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 ( $1 + 1 + 1$ ) to 27 ( $9 + 9 + 9$ )

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column =  $45/3 = 15$  [The total number of coins in each row is the same. The total number of coins in each column is also the same.] ==> (1)

Let us represent the final configuration of the sacks in boxes as follows:

|     | Table |     |     |
|-----|-------|-----|-----|
|     | C-1   | C-2 | C-3 |
| R-1 |       |     |     |
| R-2 |       |     |     |
| R-3 |       |     |     |

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also \* => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now  $c > 5$  &  $c + 8 + 9$  should be a multiple of 3 =>  $c = 7$  is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also \*\* => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also \* => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 => c = 3 for c + 18 to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

|     | Table     |           |     |
|-----|-----------|-----------|-----|
|     | C-1       | C-2       | C-3 |
| R-1 |           | 3,9,9 (7) |     |
| R-2 | 1,2,9 (4) |           |     |
| R-3 | 7,8,9 (8) |           |     |

From (1), The average in bag (1,1) is  $15 - 4 - 8 = 3$ .

From (1), The average in bag (1,3) is  $15 - 3 - 7 = 5$ .

|     | Table     |           |         |
|-----|-----------|-----------|---------|
|     | C-1       | C-2       | C-3     |
| R-1 | Avg = 3   | 3,9,9 (7) | Avg = 5 |
| R-2 | 1,2,9 (4) |           |         |
| R-3 | 7,8,9 (8) |           |         |

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and \*\* => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself  $3 \times 3 = 9$ )

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and \* => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become  $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 =>  $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) |           |           |
| R-3 | 7,8,9 (8) |           |           |

Consider bag (3,3)

0 sacks have more than 5 coins and \*\* => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now  $c = 1$  or 2 or 3 or 4 =>  $c = 1$  or 4 for number of coins to be a multiple of 3.

But  $c = 1$  as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only  $1 \times$

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | Avg = 2   | Avg = 9   |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only  $1 \times$  => smallest elements should be 1.

=> 1, b, c are the coins where  $b + c = 5$  and b,c can't be equal to 1 and less than 5 =>  $2 + 3 = 5$  is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | Avg = 6   | 1,1,1 (1) |

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and \*\* => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and  $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

|     | Table     |           |           |
|-----|-----------|-----------|-----------|
|     | C-1       | C-2       | C-3       |
| R-1 | 1,1,7 (3) | 3,9,9 (7) | 1,6,8 (5) |
| R-2 | 1,2,9 (4) | 1,2,3 (2) | 9,9,9 (9) |
| R-3 | 7,8,9 (8) | 1,8,9 (6) | 1,1,1 (1) |

Bags with different number of coins in all 3 sacks are (2,1), (3,2), (2,2), (3,2), (1,3) => 5 bags.

**Explanation [30 - 34]:**

In this set, we are told that the amount each firm raised every year increased until it reached a maximum, and then decreased until the firm closed down and no firm raised the same amount of money in two consecutive years.

The increase or decrease can be  $\pm 1$  or  $\pm 2$ .  $\Rightarrow (1)$

We are also told that each firm raised Rs. 1 crore in its first and last year of existence

Consider A:

It raised money for 8 years

$\Rightarrow$  The raising pattern looks like follows:

1, a, b, c, d, e, f, 1  $\Rightarrow$  where a, b, c, ..., f are the unknown amounts raised.

Also  $a + b + c + d + e + f = 21 - 2 = 19$ .

We can observe that  $19/6$  is slightly greater than 3  $\Rightarrow$  The average amount raised should be around 3.

If  $a = 3$  and  $f = 3 \Rightarrow b + c + d + e = 13$  (not possible) as the minimum case would be (4, 5, 6, 4)  $\Rightarrow$  Not possible.

If  $a = 3$  and  $f = 2 \Rightarrow b + c + d + e = 14$  (not possible) as the minimum case would be (4, 5, 4, 3)  $\Rightarrow$  Not possible.

$\Rightarrow a = 2$  and  $f = 2 \Rightarrow b + c + d + e = 15$  the minimum case is (3, 4, 5, 3) or (3, 5, 4, 3) which gives a sum of 15.

So, the possible cases for A are:

|   | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|------|------|------|------|------|------|------|------|
| A | 1    | 2    | 3    | 4    | 5    | 3    | 2    | 1    |
|   | 1    | 2    | 3    | 5    | 4    | 3    | 2    | 1    |

Consider B:

The patterns looks as follows:

1, a, b, 1

If  $a = 2$ , b has to be equal to 3 to satisfy (1)

if  $a = 3$ , b has to be equal to 2 to satisfy (1)

$\Rightarrow$  The possible cases for B are:

| B | 2012 | 2013 | 2014 | 2015 |
|---|------|------|------|------|
|   | 1    | 2    | 3    | 1    |
|   | 1    | 3    | 2    | 1    |

Consider C:

The pattern looks as follows:

1, ..., 1

Let us assume there are 2 gaps between  $\Rightarrow a + b = 7$  (Not possible) as maximum case would be 1, 3, 2, 1

Let us assume there are 3 gaps between  $\Rightarrow a + b + c = 7$ , the minimum case possible is 1, 2, 3, 2, 1  $\Rightarrow$  Satisfies.

Now, if there are 4 gaps  $\Rightarrow a + b + c + d = 7 \Rightarrow$  The average value is  $7/4$  which is less than 2  $\Rightarrow$  Not possible.

$\Rightarrow$  The possible cases for C are:

| C | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|------|------|------|------|------|
|   | 1    | 2    | 3    | 2    | 1    |

Consider D:

The pattern looks as follows:

1, a, b, c, 1

$\Rightarrow a + b + c = 8$

When  $a = 2$  and  $c = 2 \Rightarrow b = 4 \Rightarrow 2, 4, 2 \Rightarrow$  Satisfies.

When  $a = 2$  and  $c = 3$ , b should be 3 (Not satisfying (1))

When  $a = 3$  and  $c = 3$ , b should be 2 (Not satisfying (1))

$\Rightarrow$  The possible cases for D are:

| D | 2011 | 2012 | 2013 | 2014 | 2015 |
|---|------|------|------|------|------|
|   | 1    | 2    | 4    | 2    | 1    |

Consider E:

The pattern looks as follows:

1, ..., 1

For 1 or 2 gaps, we can't get a sum of 11.

Assume 3 gaps  $\Rightarrow a + b + c = 11$ , the maximum case is 3, 5, 3  $\Rightarrow$  Satisfies.

Now, assume 4 gaps

$\Rightarrow a + b + c + d = 11$ , the minimum case is 2, 3, 4, 2 or 2, 4, 3, 2 which satisfies (1) and  $2 + 3 + 4 + 2 = 11$ .

$\Rightarrow$  The possible cases for E are:

| E | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |
|---|------|------|------|------|------|------|
|   | 1    | 3    | 5    | 3    | 1    | -    |
|   | 1    | 2    | 3    | 4    | 2    | 1    |
|   | 1    | 2    | 4    | 3    | 2    | 1    |

In summary, the possible cases for all 5 companies is:

|   | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|---|------|------|------|------|------|------|------|------|
| A | 1    | 2    | 3    | 4    | 5    | 3    | 2    | 1    |
|   | 1    | 2    | 3    | 5    | 4    | 3    | 2    | 1    |
| B | 2012 | 2013 | 2014 | 2015 |      |      |      |      |
|   | 1    | 2    | 3    | 1    |      |      |      |      |
|   | 1    | 3    | 2    | 1    |      |      |      |      |
| C | 2013 | 2014 | 2015 | 2016 | 2017 |      |      |      |
|   | 1    | 2    | 3    | 2    | 1    |      |      |      |
| D | 2011 | 2012 | 2013 | 2014 | 2015 |      |      |      |
|   | 1    | 2    | 4    | 2    | 1    |      |      |      |
| E | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 |      |      |
|   | 1    | 3    | 5    | 3    | 1    | -    |      |      |
|   | 1    | 2    | 3    | 4    | 2    | 1    |      |      |
|   | 1    | 2    | 4    | 3    | 2    | 1    |      |      |

30. B

We see that only for C and D, we can conclude the amounts raised with certainty.

[VIDEO SOLUTION](#)



31. C

Money raised in 2015 is  $2 + 1 + 3 + 1 + 0/1 = 7$  or 8.

[VIDEO SOLUTION](#)

32. 17

Maximum money raised in 2013 is  $5 + 3 + 1 + 4 + 4 = 17$ .

[VIDEO SOLUTION](#)

33. C

Given that E raised 3 in 2013 => in 2012 he could have raised a minimum of 4 crores.

=> Minimum amount is  $4 + 1 + 0 + 2 + 4 = 11$ .

[VIDEO SOLUTION](#)

34. A

Given that total amount raised in 2014 is 12

=>  $3 + 3/2 + 2 + 2 + 1/2 = 12$  =>

=> possible case is  $3 + 3 + 2 + 2 + 2 = 12$ .

A) In 2013, B raised 2 crores and E also raised 3/4 crores => Not Possible.

B) In 2013, A could have raised 5/4 and D raised 4 => Possible.

C) In 2014, A raised 3 and B raised 3 => Possible.

D) In 2014, B raised 3 where as E raised 2 =>  $3 > 2$  => Possible.

[VIDEO SOLUTION](#)

35. B

| Table 1: 2-day averages for Days 2 through 5 |       |       |       |
|----------------------------------------------|-------|-------|-------|
| Day 2                                        | Day 3 | Day 4 | Day 5 |
| 15                                           | 15.5  | 16    | 17    |

Let the total score of day 1, day 2, day 3, day 4, and day 5 are  $d_1, d_2, d_3, d_4$ , and  $d_5$ , respectively.

The table shows that  $d_1 + d_2 = 30$  ... eq (1),  $d_2 + d_3 = 31$  ... eq (2),  $d_3 + d_4 = 32$  .... eq(3),  $d_4 + d_5 = 34$  ... eq(4)

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore,  $d_3 = d_4$  =>  $d_3 = d_4 = 16$ , which implies  $d_2 = 15$ ,  $d_5 = 18$ , and  $d_1 = 15$ .

The day-wise score is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       |       |       |
| Bimal       |       |       |       |       |       |
| Chatur      |       |       |       |       |       |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       | 3     |       |
| Bimal       |       |       |       | 7     |       |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be  $x$ . Therefore,  $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       | 5     | 3     | 5/4   |
| Bimal       |       |       | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be  $a$ , and the score of Bimal be  $b$ . Then  $9+a+b = 15$ , and  $a > b$

$\Rightarrow a+b=6$ , and  $a > b$

Hence, the value of  $a$  can be  $4/5$ , and the value of  $b$  can be  $2/1$

Therefore, the final table is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       | 7     | 4/5   | 5     | 3     | 5/4   |
| Bimal       | 5     | 2/1   | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From the table, we can see that the score of Akhil is 7 on day 1.

The correct option is B

[VIDEO SOLUTION](#)



36.D

| Table 1: 2-day averages for Days 2 through 5 |       |       |       |
|----------------------------------------------|-------|-------|-------|
| Day 2                                        | Day 3 | Day 4 | Day 5 |
| 15                                           | 15.5  | 16    | 17    |

Let the total score of day 1, day 2, day 3, day 4, and day 5 are  $d_1$ ,  $d_2$ ,  $d_3$ ,  $d_4$ , and  $d_5$ , respectively.

The table shows that  $d_1+d_2 = 30 \dots \text{eq (1)}$ ,  $d_2+d_3 = 31 \dots \text{eq (2)}$ ,  $d_3+d_4 = 32 \dots \text{eq(3)}$ ,  $d_4+d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore,  $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$ , which implies  $d_2 = 15$ ,  $d_5 = 18$ , and  $d_1 = 15$ .

The day-wise score is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       |       |       |
| Bimal       |       |       |       |       |       |
| Chatur      |       |       |       |       |       |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       | 3     |       |
| Bimal       |       |       |       | 7     |       |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be  $x$ . Therefore,  $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       | 5     | 3     | 5/4   |
| Bimal       |       |       | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil's on Day 2.

Let the score of Akhil be  $a$ , and the score of Bimal be  $b$ . Then  $9+a+b = 15$ , and  $a > b$

$\Rightarrow a+b = 6$ , and  $a > b$

Hence, the value of  $a$  can be  $4/5$ , and the value of  $b$  can be  $2/1$

Therefore, the final table is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Total score |
|-------------|-------|-------|-------|-------|-------|-------------|
| Akhil       | 7     | 4/5   | 5     | 3     | 5/4   | 23/24/25    |
| Bimal       | 5     | 2/1   | 5     | 7     | 7/8   | 27/26/25    |
| Chatur      | 3     | 9     | 6     | 6     | 6     | 30          |
| Total Score | 15    | 15    | 16    | 16    | 18    | 80          |

From the table, we can see that the maximum score is obtained by Chatur.

The correct option is D

[VIDEO SOLUTION](#)

37.25

| Table 1: 2-day averages for Days 2 through 5 |       |       |       |
|----------------------------------------------|-------|-------|-------|
| Day 2                                        | Day 3 | Day 4 | Day 5 |
| 15                                           | 15.5  | 16    | 17    |

Let the total score of day 1, day 2, day 3, day 4, and day 5 are  $d_1$ ,  $d_2$ ,  $d_3$ ,  $d_4$ , and  $d_5$ , respectively.

The table shows that  $d_1+d_2 = 30 \dots$  eq (1),  $d_2+d_3 = 31 \dots$  eq (2),  $d_3+d_4 = 32 \dots$  eq(3),  $d_4+d_5 = 34 \dots$  eq(4)

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore,  $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$ , which implies  $d_2 = 15$ ,  $d_5 = 18$ , and  $d_1 = 15$ .

The day-wise score is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       |       |       |
| Bimal       |       |       |       |       |       |
| Chatur      |       |       |       |       |       |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       | 3     |       |
| Bimal       |       |       |       | 7     |       |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be  $x$ . Therefore,  $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       | 5     | 3     | 5/4   |
| Bimal       |       |       | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be  $a$ , and the score of Bimal be  $b$ . Then  $9+a+b = 15$ , and  $a > b$

$\Rightarrow a+b=6$ , and  $a > b$

Hence, the value of  $a$  can be  $4/5$ , and the value of  $b$  can be  $2/1$

Therefore, the final table is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Total score |
|-------------|-------|-------|-------|-------|-------|-------------|
| Akhil       | 7     | 4/5   | 5     | 3     | 5/4   | 23/24/25    |
| Bimal       | 5     | 2/1   | 5     | 7     | 7/8   | 27/26/25    |
| Chatur      | 3     | 9     | 6     | 6     | 6     | 30          |
| Total Score | 15    | 15    | 16    | 16    | 18    | 80          |

From the table, we can see that the minimum score obtained by Bimal is 25.

[VIDEO SOLUTION](#)

38.D

| Table 1: 2-day averages for Days 2 through 5 |       |       |       |
|----------------------------------------------|-------|-------|-------|
| Day 2                                        | Day 3 | Day 4 | Day 5 |
| 15                                           | 15.5  | 16    | 17    |

Let the total score of day 1, day 2, day 3, day 4, and day 5 are  $d_1$ ,  $d_2$ ,  $d_3$ ,  $d_4$ , and  $d_5$ , respectively.

The table shows that  $d_1+d_2 = 30 \dots \text{eq (1)}$ ,  $d_2+d_3 = 31 \dots \text{eq (2)}$ ,  $d_3+d_4 = 32 \dots \text{eq(3)}$ ,  $d_4+d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore,  $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$ , which implies  $d_2 = 15$ ,  $d_5 = 18$ , and  $d_1 = 15$ .

The day-wise score is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       |       |       |
| Bimal       |       |       |       |       |       |
| Chatur      |       |       |       |       |       |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       | 3     |       |
| Bimal       |       |       |       | 7     |       |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be  $x$ . Therefore,  $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       | 5     | 3     | 5/4   |
| Bimal       |       |       | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be  $a$ , and the score of Bimal be  $b$ . Then  $9+a+b = 15$ , and  $a > b$

$\Rightarrow a+b = 6$ , and  $a > b$

Hence, the value of  $a$  can be  $4/5$ , and the value of  $b$  can be  $2/1$

Therefore, the final table is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Total score |
|-------------|-------|-------|-------|-------|-------|-------------|
| Akhil       | 7     | 4/5   | 5     | 3     | 5/4   | 23/24/25    |
| Bimal       | 5     | 2/1   | 5     | 7     | 7/8   | 27/26/25    |
| Chatur      | 3     | 9     | 6     | 6     | 6     | 30          |
| Total Score | 15    | 15    | 16    | 16    | 18    | 80          |

In the question, it is given that the total score obtained by Bimal is a multiple of 3, which implies the total score obtained by Bimal is 27, which implies the total score obtained by Akhil is 23.

Akhil will score 23, when his scores on Days 1, 2, 3, 4, and 5 are 7, 4, 5, 3, 4, respectively.

Hence, the score obtained by him on Day 2 is 4.

The correct option is D

[VIDEO SOLUTION](#)

39.26

| Table 1: 2-day averages for Days 2 through 5 |       |       |       |
|----------------------------------------------|-------|-------|-------|
| Day 2                                        | Day 3 | Day 4 | Day 5 |
| 15                                           | 15.5  | 16    | 17    |

Let the total score of day 1, day 2, day 3, day 4, and day 5 are  $d_1$ ,  $d_2$ ,  $d_3$ ,  $d_4$ , and  $d_5$ , respectively.

The table shows that  $d_1+d_2 = 30 \dots \text{eq (1)}$ ,  $d_2+d_3 = 31 \dots \text{eq (2)}$ ,  $d_3+d_4 = 32 \dots \text{eq(3)}$ ,  $d_4+d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore,  $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$ , which implies  $d_2 = 15$ ,  $d_5 = 18$ , and  $d_1 = 15$ .

The day-wise score is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       |       |       |
| Bimal       |       |       |       |       |       |
| Chatur      |       |       |       |       |       |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       |       | 3     |       |
| Bimal       |       |       |       | 7     |       |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be  $x$ . Therefore,  $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|-------------|-------|-------|-------|-------|-------|
| Akhil       |       |       | 5     | 3     | 5/4   |
| Bimal       |       |       | 5     | 7     | 7/8   |
| Chatur      | 3     | 9     | 6     | 6     | 6     |
| Total Score | 15    | 15    | 16    | 16    | 18    |

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be  $a$ , and the score of Bimal be  $b$ . Then  $9+a+b = 15$ , and  $a > b$

$$\Rightarrow a+b=6, \text{ and } a>b$$

Hence, the value of a can be 4/5, and the value of b can be 2/1

Therefore, the final table is given below:

|             | Day 1 | Day 2 | Day 3 | Day 4 | Day 5 | Total score |
|-------------|-------|-------|-------|-------|-------|-------------|
| Akhil       | 7     | 4/5   | 5     | 3     | 5/4   | 23/24/25    |
| Bimal       | 5     | 2/1   | 5     | 7     | 7/8   | 27/26/25    |
| Chatur      | 3     | 9     | 6     | 6     | 6     | 30          |
| Total Score | 15    | 15    | 16    | 16    | 18    | 80          |

In the question, it is given that the score obtained by Akhil is 24, which implies the score obtained by Bimal is 26.

The answer is 26

[VIDEO SOLUTION](#)

#### Explanation [40 - 44]:

Consider Statement 2: Anjali took Ride-1 at 11 am after waiting for 30 minutes for Chitra to complete it. It was the only ride where Anjali waited.

This implies that Chitra took Ride 1 at 10 am. Now we also know that she spent Rs 50 and that she left at 11 am. Now, since she did one ride costing Rs 20 at 10, she must have taken Ride-3 at 9 am.

So we get the following table for Chitra.

|      | Ride 3      | Ride 1       |
|------|-------------|--------------|
| Time | 9 am- 10 am | 10 am- 11 am |
| Cost | Rs 30       | Rs 20        |

Now we know that Chitra and Anjali spent Rs 50 before 12:15 pm. It is not possible for Anjali to go on Ride-3 at 10 am as we know that she was waiting for 30 minutes before taking Ride-1 (She was waiting from 10:30 am).

Now, since we know that Ride-1 was the only ride for which she waited, we can say that she took Ride-1 at 11 am and started Ride-3 at 12 am

So we get the following table for Anjali.

|      | Ride-1       | Ride-3      |
|------|--------------|-------------|
| Time | 11 am- 12 pm | 12 pm - 1pm |
| Cost | 20           | 30          |

Now, we know that Bipasha started her first ride at 11:30 am. We also know that they all spent Rs 50 before 12:15 pm.

Therefore, the first ride Bipasha takes will be Ride-2, costing Rs 50.

So we get the following table for Bipasha.

|      | Ride-2            |  |
|------|-------------------|--|
| Time | 11:30 am- 12:30am |  |
| Cost | 50                |  |

We know that Ride 3 stops at 1 pm. So the last ride taken by Anjali will either be Ride-2 or Ride-4. Now, considering Statement 4, we know that the last ride taken by Anjali and Bipasha was same and that Bipasha rode it after Anjali. So their last ride can't be 2.

So the last ride of both Bipasha and Anjali will be 4.

Now if we assume that immediately after ending Ride-3, Anjali goes to Ride-4, then the last ride of Bipasha will be Ride-4 from 2 pm - 3 pm. But we know that Bipasha rode 3 rides. So this case is not possible.

Since Anjali didn't have any break or waiting time, the only ride she can ride at 1 pm will be Ride 2 and then she will go on Ride-4 from 2 pm to 3 pm.

So we get the following table for Anjali:

|      | Ride-1       | Ride-3       | Ride-2     | Ride-4    |
|------|--------------|--------------|------------|-----------|
| Time | 11 am- 12 pm | 12 pm - 1 pm | 1 pm- 2 pm | 2 pm- 3pm |
| Cost | 20           | 30           | 50         | 40        |

Now we know that the last ride that Bipasha took was Ride-4 and that she had a gap of 1.5 hrs before it. This is only possible when she takes one ride between Ride-2 and Ride-4. Since Ride-3 is closed at 1 pm, she can only take Ride 1. So we get the following table for her.

|      | Ride-2            | Ride-1            | Break            | Waiting time      | Ride-4      |
|------|-------------------|-------------------|------------------|-------------------|-------------|
| Time | 11:30 am-12:30 pm | 12:30 pm- 1:30 pm | 1:30 pm- 2:30 pm | 2:30 pm to 3:00pm | 3 pm - 4 pm |
| Cost | 50                | 20                |                  |                   | 40          |

40. C

As we can see from the table for Bipasha, she spent a total of 50+20+40= 110

Therefore the required answer is Option C: 110

[VIDEO SOLUTION](#)



41. B

Anjali completed a total of 4 rides, 3 of which were completed at 2. Therefore the answer is Option B: Ride-1, Ride 3, and Ride -2

[VIDEO SOLUTION](#)

42. A

Only Ride-1 was taken by all the visitors. Therefore the correct answer is Option A: Ride-1

VIDEO SOLUTION

43.6

Anjali took 4 rides, and Chitra took 2 rides. Therefore the correct answer is 6

VIDEO SOLUTION

44.140

As we can see from the table of Anjali she spent a total of  $20+30+50+40=140$

Therefore the required answer is 140

VIDEO SOLUTION

Quant

45.D

It is given that  $a^m \cdot b^n = 144^{145}$ , where  $a > 1$  and  $b > 1$ .

144 can be written as  $144 = 2^4 \times 3^2$

Hence,  $a^m \cdot b^n = 144^{145}$  can be written as  $a^m \cdot b^n = (2^4 \times 3^2)^{145} = 2^{580} \times 3^{290}$

We know that  $3^{290}$  is a natural number, which implies it can be written as  $a^1$ , where  $a > 1$

Hence, the least possible value of m is 1. Similarly, the largest value of n is 580.

Hence, the largest value of (n-m) is  $(580-1) = 579$

The correct option is D

VIDEO SOLUTION



46.B

It is given that  $\frac{x}{y} < \frac{x+3}{y-3}$ , which can be written as  $\frac{x}{y} - \frac{x+3}{y-3} < 0$

$$\Rightarrow \frac{x(y-3)-y(x+3)}{y(y-3)} < 0$$

$$\Rightarrow \frac{xy-3x-xy-3y}{y(y-3)} < 0$$

$$\Rightarrow \frac{-3(x+y)}{y(y-3)} < 0$$

$$\Rightarrow \frac{3(x+y)}{y(y-3)} > 0$$

From this inequality, we can say that, when  $y < 0 \Rightarrow y(y-3) > 0$ . Now to satisfy the given equation  $\frac{3(x+y)}{y(y-3)} > 0$ ,

$(x+y)$  must be greater than zero Hence,  $x > 0$  and  $|x| > |y|$

Therefore, the magnitude of  $x$  is greater than the magnitude of  $y$ .

Hence,  $x > y$ , and  $|x| > |y| \Rightarrow -x < y$  (Since the magnitude of  $x$  is greater than the magnitude of  $y$ .)

The correct option is B.

VIDEO SOLUTION

47.C

It is given that k divides  $m+2n$  and  $3m+4n$ .

Since k divides  $(m+2n)$ , it implies k will also divide  $3(m+2n)$ . Therefore, k divides  $3m+6n$ .

Similarly, we know that k divides  $3m+4n$ .

We know that if two numbers a, and b both are divisible by c, then their difference (a-b) is also divisible by c.

By the same logic, we can say that  $\{(3m+6n)-(3m+4n)\}$  is divisible by k. Hence,  $2n$  is also divisible by k.

Now,  $(m+2n)$  is divisible by k, it implies  $2(m+2n)=2m+4n$  is also divisible by k.

Hence,  $\{(3m+4n)-(2m+4n)\} = m$  is also divisible by k.

Therefore, m, and  $2n$  are also divisible by k.

The correct option is C

VIDEO SOLUTION

48.D

It is given that  $2^{4x^2} - 2^{2x^2+x+16} + 2^{2x+30} = 0$ , which can be written as:

$$\Rightarrow (2^{2x^2})^2 - 2^{2x^2} \cdot 2^{x+15} \cdot 2^1 + (2^{x+15})^2 = 0$$

$$\Rightarrow (2^{2x^2} - 2^{x+15})^2 = 0$$

$$\Rightarrow 2^{2x^2} - 2^{x+15} = 0 \text{ (Since } (a-b)^2 = 0 \Rightarrow a-b=0)$$

$$\Rightarrow 2x^2 = x + 15$$

$$\Rightarrow 2x^2 - x - 15 = 0$$

$$\Rightarrow 2x^2 - 6x + 5x - 15 = 0$$

$$\Rightarrow 2x(x-3) + 5(x-3) = 0$$

$$\Rightarrow (2x+5)(x-3) = 0$$

Hence, the possible values of  $x$  are  $-\frac{5}{2}$ , and 3, respectively.

Therefore, the sum of the possible values is  $(3 - \frac{5}{2}) = \frac{1}{2}$

The correct option is D

[VIDEO SOLUTION](#)

49.15

Since there are two distinct factors other than 1, and itself, which implies the total number of factors of  $N$  is 4.

It can be done in two ways.

First case:  $N = p^3$  (where  $p$  is a prime number)

Second case:  $N = p_1 \times p_2$  (Where  $p_1, p_2$  are the prime numbers)

From case 1, we can see that the numbers which is a cube of prime and less than 50 are 8, and 27 (2 numbers).

From case 2, we will get the numbers in the form  $(2*3), (2*5), (2*7), (2*11), (2*13), (2*17), (2*19), (2*23), (3*5), (3*7), (3*11), (3*13), (5*7)$  (13 numbers)

Hence, the total number of numbers having two distinct factors is  $(13+2) = 15$ .

[VIDEO SOLUTION](#)

50.7

It is given that  $\log_{\sqrt{3}}(x) + \frac{\log_x(25)}{\log_x(0.008)} = \frac{16}{3}$ , which can be written as:

$$\Rightarrow 2 \log_3 x + \log_{0.008} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{8}{1000}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{\frac{1}{125}} 25 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x + \log_{5^{-3}} (5)^2 = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x - \frac{2}{3} = \frac{16}{3}$$

$$\Rightarrow 2 \log_3 x = \frac{16}{3} + \frac{2}{3}$$

$$\Rightarrow 2 \log_3 x = 6$$

$$\Rightarrow \log_3 x^2 = 6 \Rightarrow x^2 = 3^6$$

$$\text{Hence, } \log_3 (3 \cdot x^2) = \log_3 (3 \cdot 3^6) = \log_3 3^7 = 7$$

[VIDEO SOLUTION](#)

## 100+ CAT Quant Questions



51.6

It is given that  $(x-1)^2 + 2kx + 11 = 0$  has no real roots. (Where  $k$  is the largest integer)

$(x-1)^2 + 2kx + 11 = 0$ , which can be written as:

$$\Rightarrow x^2 - 2x + 1 + 2kx + 11 = 0$$

$$\Rightarrow x^2 - 2(k-1)x + 12 = 0$$

We know that for no real roots,  $D < 0 \Rightarrow b^2 - 4ac < 0$

$$\text{Hence, } \{2(k-1)\}^2 - 4 \cdot 1 \cdot 12 < 0$$

$$\Rightarrow 4(k-1)^2 < 48$$

$$\Rightarrow (k-1)^2 < 12$$

Since  $k$  is an integer, it implies  $(k-1)$  is also an integer.

Therefore, from the above inequality, we can say that the largest possible value of  $(k-1) = 3 \Rightarrow$  The largest possible value of  $k$  is 4.

Now we need to calculate the least possible value of  $\frac{k}{4y} + 9y$ .

$$\frac{k}{4y} + 9y \text{ can be written as } \frac{4}{4y} + 9y = \frac{1}{y} + 9y$$

The least possible value of  $9y + \frac{1}{y}$  can be calculated using A.M-G.M inequality.

Using A.M-G.M inequality, we get:

$$\frac{9y + \frac{1}{y}}{2} \geq \sqrt{9y \times \frac{1}{y}}$$

$$\Rightarrow \frac{9y + \frac{1}{y}}{2} \geq \sqrt{9}$$

$$\Rightarrow \frac{9y + \frac{1}{y}}{2} \geq 3$$

$$\Rightarrow 9y + \frac{1}{y} \geq 6$$

Hence, the least possible value is 6

[VIDEO SOLUTION](#)

52. A

Let the time taken by A to fill the tank alone be  $x$  hours, which implies the time taken by B to empty the tank alone is  $(x-1)$  hours (B is the drainage pipe), and the time taken by C to fill the tank is  $y$  hours.

It is given that when pipes A, B, and C are turned on together, the empty tank is filled in two hours.

$$\text{Hence, } \frac{1}{x} - \frac{1}{x-1} + \frac{1}{y} = \frac{1}{2} \dots \text{Eq(1)}$$

It is given that if pipes B and C are turned on together when the tank is empty and Pipe B is turned off after one hour, then Pipe C takes another one hour and 15 minutes to fill the remaining tank.

Hence, B worked for 1 hour, and C worked for 2 hours 15 minutes, which is equal to  $\frac{9}{4}$  hours.

In 1 hour, B worked  $-\frac{1}{x-1}$  units, and in  $\frac{9}{4}$  hours, C worked  $\frac{9}{4y}$  units.

$$\text{Hence, } \frac{9}{4y} - \frac{1}{x-1} = 1 \dots \text{Eq(2)}$$

Solving both equations, we get  $y = \frac{3}{2}$ , and  $x = 3$

Hence, the time taken by C is  $\frac{3}{2}$  hours, which is equal to 90 minutes.

The correct option is A

[VIDEO SOLUTION](#)

53. B

It is given that Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. It is also known that he repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year.

$$\text{The total amount at the end of the first year is: } 200000 \times \frac{104}{100} \times \frac{104}{100} = 216320$$

He repays 10320 rupees at the end of the first year, which implies the amount that remains unpaid at the end of the first year is 206000 rupees.

This unpaid amount will accrue interest for another two years.

$$\text{Hence, the final amount at the end of three years is } 206000 \times \frac{104}{100} \times \frac{104}{100} \times \frac{104}{100} \times \frac{104}{100} = 240990.86$$

Hence, the accrued interest in these two years is  $(240990.86 - 206000) = 34990.86$  rupees.

Hence, the total interest accrued over the three years =  $(34990.86 + 16320) = 51311$  rupees.

The correct option is B

[VIDEO SOLUTION](#)

54. C

It is given that the speed of Ravi is 40 kmph, which is equal to  $\frac{100}{9}$  m/s. It is also known that the speed of Ashok is 50 kmph, which is equal to  $\frac{125}{9}$  m/s.

It is known that the distance between Ravi and Ashok is 225 meters, and the relative speed of Ravi and Ashok is  $\frac{125}{9} + \frac{100}{9} = 25$  m/s

Hence, they will meet each other in  $\frac{225}{25} = 9$  seconds. The distance traveled by Ravi in these 9 seconds is  $\frac{100}{9} \times 9 = 100$  meters.

Since Vijay was already 54 meters behind Ravi when they were starting, Vijay must travel  $(100 + 54) = 154$  meters in these 9 seconds.

Hence, the speed of Vijay is  $\frac{154}{9}$  m/s, which is equal to  $\frac{154}{9} \times \frac{18}{5} = \frac{308}{5} = 61.6$  kmph.

The correct option is C

[VIDEO SOLUTION](#)

55. C

The cost price of the sunglass for Meenu when he purchased it for the first time was 1000 rupees, and he sold it to Kanu at 20% profit. Hence, the selling price of the sunglass is 1200 rupees, which Kanu purchased. Hence, the profit made by Meenu is  $(1200 - 1000) = 200$  rupees.

Hence, the cost price of the same sunglass for Kanu is 1200 rupees, and now he sold it to Meenu at a 20% loss. Hence, the selling price of the sunglass now is  $(1200 \times 0.8) = 960$  rupees.

The cost price of the same sunglass for Meenu when he purchased it for the second time was 960 rupees. Now Meenu sold it Tanu, at a certain price such that the total profit of Meenu becomes 500 rupees.

Hence, on the second transaction (selling it to Tanu), Meenu made a profit of  $(500 - 200) = 300$  rupees.

Hence, the profit made by Minu in the second transaction is  $(300/960) \times 100\% = 31.25\%$

The correct option is C

[VIDEO SOLUTION](#)

**100+ CAT DILR Questions**



56. D

it is given that the price of a precious stone is directly proportional to the square of its weight. Let the price be denoted by C and the weight is denoted by W.

Hence,  $C \propto W^2 \Rightarrow C = kw^2$  (where k is the proportional constant)

Now, Sita has a precious stone weighing 18 units.

Therefore,  $C = kw^2 = k \cdot 18^2 = 324$

If she breaks it into four pieces with each piece having a distinct integer weight, then the difference between the highest and lowest possible values of the total price of the four pieces will be 288000.

To get the lowest possible value of C, we will get the weight of the four-piece as close as possible (3,4,5,6). To get the highest value we will try to take three pieces as low as possible, and one is as high as possible (1, 2, 3, 12).

Hence, the maximum cost =  $k(12^2 + 1^2 + 2^2 + 3^2) = 158k$ , and the minimum cost is  $k(3^2 + 4^2 + 5^2 + 6^2) = 86k$

Hence, the difference is  $(158k - 86k) = 72k$ , which is equal to 288000.

$\Rightarrow 72k = 288000$

$\Rightarrow k = 4000$

Hence, the price of the original stone is  $324k = 324 \times 4000 = 1296000$

The correct option is D

VIDEO SOLUTION

57. B

Let the number of total employees in the company be 100x, and the total salary of all the employees be 100y.

It is given that 20% of the employees work in the manufacturing department, and the total salary obtained by all the manufacturing employees is one-sixth of the total salary obtained by all the employees in the company.

Hence, the total number of employees in the manufacturing department is 20x, and the total salary received by them is  $(100y/6)$

Average salary in the manufacturing department =  $(100y/6 \cdot 20x) = 5y/6x$

Similarly, the total number of employees in the nonmanufacturing department is 80x, and the total salary received by them is  $(500y/6)$

Hence, the average salary in the nonmanufacturing department =  $(500y/6 \cdot 80x) = 25y/24x$

Hence, the ratio is:-  $(5y/6x) : (25y/24x)$

$\Rightarrow 120 : 150 = 4 : 5$

The correct option is B

VIDEO SOLUTION

58. 16

It is given that if a certain amount of money is divided equally among n persons, each one receives Rs 352. Hence, the total amount of money is  $(352 \cdot n) = 352n$  ... Eq(1)

It is also known that if two persons receive Rs 506 each and the remaining amount is divided equally among the other persons, each of them receives less than or equal to Rs 330

Hence, the maximum amount of money with them =  $506 \cdot 2 + (n-2) \cdot 330 = 1012 + 330n - 660 = 352 + 330n$

Now,  $352 + 330n \geq 352n$

$\Rightarrow 22n \leq 352$

$\Rightarrow n \leq 16$

Hence, the maximum value is 16

VIDEO SOLUTION

59. 407

Let the number of white shirts be m, and the number of blue shirts be n. Hence, the total cost of the shirts =  $(1000m + 1125n)$ , and the number of shirts is  $(m+n)$

The average price of the shirts is  $\frac{1000m + 1125n}{m+n}$ .

It is given that he set a fixed market price which was 25% higher than the average cost of all the shirts. He sold all the shirts at a discount of 10%.

Hence, the average selling price of the shirts =  $\left( \frac{1000m + 1125n}{m+n} \right) \times \frac{5}{4} \times \frac{9}{10} = \frac{9}{8} \left( \frac{1000m + 1125n}{m+n} \right)$

The average profit of the shirts =  $\frac{9}{8} \left( \frac{1000m + 1125n}{m+n} \right) - \frac{1000m + 1125n}{m+n} = \frac{1}{8} \left( \frac{1000m + 1125n}{m+n} \right)$

The total profit of the shirts =  $\frac{1}{8} \left( \frac{1000m + 1125n}{m+n} \right) \times (m+n) = \frac{1}{8} (1000m + 1125n)$

Now,  $\Rightarrow \frac{1}{8} (1000m + 1125n) = 51000$

$\Rightarrow 1000m + 1125n = 51000 \times 8 = 408000$

Now to get the maximum number of shirts, we need to minimize n (since the coefficient of n is greater than the coefficient of m), but it can't be zero. Therefore, m has to be maximum.

$$m = \frac{408000 - 1125n}{1000}$$

The maximum value of m such that m, and both are integers is  $m = 399$ , and  $n = 8$  (by inspection)

Hence, the maximum number of shirts =  $m+n = 399+8 = 407$

VIDEO SOLUTION

60.7

Let's assume that after  $n$  iteration, the volume of the milk will be less than 50%, which is less than 20 liters.

Initially, the amount of milk is 40 liters, after the first iteration, the volume of milk is  $40 \cdot \frac{9}{10}$

After the second iteration, the volume of milk is  $40 \times \left(\frac{9}{10}\right)^2$

Similarly, after the  $n$  iterations, the volume of milk is  $40 \times \left(\frac{9}{10}\right)^n$

Now,

$$40 \times \left(\frac{9}{10}\right)^n \leq 20$$

$$\Rightarrow \left(\frac{9}{10}\right)^n \leq \frac{1}{2}$$

$$\Rightarrow n \geq 7$$

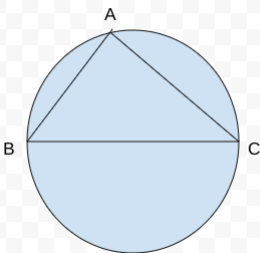
Hence, the correct answer is 7

[VIDEO SOLUTION](#)

## 100+ CAT VARC Questions



61. B



Since BC is the diameter of the circle, which implies angle BAC is 90 degrees. Let  $AB = a$  cm, which implies  $AC = b$  cm. Hence,  $BC = \sqrt{a^2 + b^2}$ , which is diameter of the circle ( $2r$ ).

$$\text{Hence, } 2r = \sqrt{a^2 + b^2}$$

$$\Rightarrow 4r^2 = a^2 + b^2$$

The area of the triangle is  $\frac{1}{2} \times a \times b$ , which can be written as

$$\Rightarrow \frac{a \cdot b}{2(a^2 + b^2)} \times (a^2 + b^2)$$

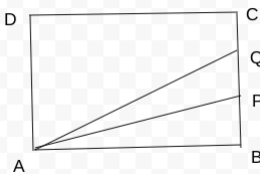
$$\Rightarrow \frac{a \cdot b}{2(a^2 + b^2)} \times 4r^2$$

$$\Rightarrow \frac{a \cdot b}{a^2 + b^2} \times 2r^2$$

The correct option is B

[VIDEO SOLUTION](#)

62. C

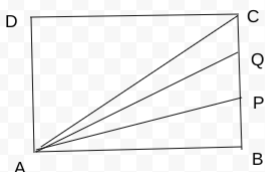


It is given that  $AB = 9$  cm,  $BC = 6$  cm.

It is also known that the areas of the figures ABP, APQ, and AQCD are in geometric progression.

Hence, the area of the ABP, APQ, and AQCD are  $k$ ,  $2k$ , and  $4k$  respectively.

The ratio of BP, PQ, QC will be the ratio of the respective triangles. Hence, we can draw a line from point A to point C.



Let the area of triangle AQC be  $x$ , which implies the area of triangle ADC =  $ADQC - AQC = 4k - x$ , which is equal to the sum of the area of triangle APB, APQ, and ACQ, respectively.

$$\text{Therefore, } 4k - x = 3k + x$$

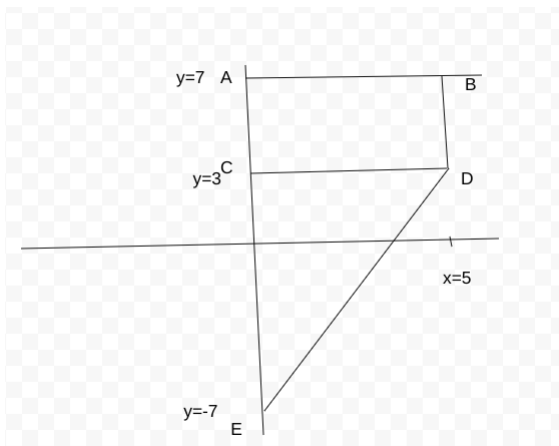
$$\Rightarrow x = k/2$$

Hence the ratio of BP: PQ: CQ = k:2k: k/2 = 2:4:1

[VIDEO SOLUTION](#)

63.45

From the inequality and nature of x, and y, we get the given diagram:



We need to find the area of the quadrilateral ABDE = area of rectangle ABCD + area of triangle CDE

$\Rightarrow$  Area of ABCD =  $(7-3) \times 5 = 20$  units, and the area of triangle CDE =  $(1/2) \times 10 \times 5 = 25$  units.

Hence, the area of the quadrilateral ABDE =  $(20+25) = 45$  units.

[VIDEO SOLUTION](#)

64. B

Let the first term of both series be  $a_1$ , and  $b_1$ , respectively, and the common difference be  $d_1$ , and  $d_2$ , respectively.

It is given that  $a_5 = b_9$ , which implies  $a_1 + 4d_1 = b_1 + 8d_2$

$$\Rightarrow a_1 - b_1 = 8d_2 - 4d_1 \dots \text{Eq(1)}$$

Similarly, it is known that  $a_{19} = b_{19}$ , which implies  $a_1 + 18d_1 = b_1 + 18d_2$

$$\Rightarrow a_1 - b_1 = 18d_2 - 18d_1 \dots \text{Eq(2)}$$

Equating (1) and (2), we get:

$$\Rightarrow 18d_2 - 18d_1 = 8d_2 - 4d_1$$

$$\Rightarrow 10d_2 = 14d_1$$

$$\Rightarrow 5d_2 = 7d_1$$

Since,  $d_1$ ,  $d_2$  are the prime numbers, which implies  $d_1 = 5$ ,  $d_2 = 7$ .

It is also known that  $b_2 = 0$ , which implies  $b_1 + d_2 = 0 \Rightarrow b_1 = -d_2 = -7$

Putting the value of  $b_1$ ,  $d_1$ , and,  $d_2$  in Eq(1), we get:

$$a_1 = 8d_2 - 4d_1 + b_1 = 56 - 20 - 7 = 29$$

$$\text{Hence, } a_{11} = a_1 + 10d_1 = 29 + 10 \cdot 5 = 29 + 50 = 79$$

The correct option is B

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65. C

Given that  $2pq - 20 = 52 - 2pq \Rightarrow 4pq = 72 \Rightarrow pq = 18 \dots (1)$

$$\text{Now, } p^2 + q^2 - 29 = 2pq - 20 \Rightarrow p^2 + q^2 - 2pq = 9 \Rightarrow (p - q)^2 = 9 \Rightarrow p - q = \pm 3$$

$$\text{Also, } p^2 + q^2 - 29 = 2pq - 20 \Rightarrow p^2 + q^2 = 2pq + 9 = 2(18) + 9 = 45$$

$$\text{Now, } p^3 - q^3 = (p - q)(p^2 + pq + q^2) = (p - q)(45 + 18) = (p - q)(63)$$

$\Rightarrow$  When  $p - q = -3 \Rightarrow$  The value is  $63(-3) = -189$  and when  $p - q = 3 \Rightarrow$  The value is  $63(3) = 189$ .

$\Rightarrow$  The difference =  $189 - (-189) = 378$ .

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## CAT Expected Questions PDF

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66. 967

It is given that  $a_n = 13 + 6(n - 1)$ , which can be written as  $a_n = 13 + 6n - 6 = 7 + 6n$

Similarly,  $b_n = 15 + 7(n - 1)$ , which can be written as  $b_n = 15 + 7n - 7 = 8 + 7n$

The common differences are 6, and 7, respectively, The common difference of terms that exists in both series is l.c.m (6, 7) = 42

The first common term of the first two series is 43 (by inspection)

Hence, we need to find the  $m$ th term, which is less than 1000, and the largest three-digit integer, and exists in both series.

$$t_m = a + (m - 1) d < 1000$$

$$\Rightarrow 43 + (m - 1) 42 < 1000$$

$$\Rightarrow (m - 1) 42 < 957$$

$$\Rightarrow m - 1 < 22.8 \Rightarrow m < 23.8 \Rightarrow m = 23$$

$$\text{Hence, the 23rd term is } 43 + 22 \times 42 = 967$$

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